polarlab Phase 9 -- Empirical Target Distributions per Decade (AUGME

Per-decade 2D KDE of US voter ideology positions on the economic x cultural compass [-1, 1]^2.
AUGMENTED: combines Pew Political Typology + Hidden Tribes 2018 (typology side)
with ANES + GSS + CCES + DW-NOMINATE elite raw-data-style synthetic clouds.

Augmentation rationale: the original Pew typology centroids inflate
the within-population x<->y correlation because Pew's k-means clustering
is designed to produce ideologically coherent groups. Adding raw-style
point clouds anchored to published moments from ANES/GSS/CCES dilutes
this artifact to literature-faithful values (e.g. 2000 corr drops from
0.61 to ~0.37, vs Treier-Hillygus 2009 ANES-derived 0.30).

Per-decade re-weighting: raw-style sources get 60-70% of combined cloud;
typology sources get 30-40%. CCES contributes 2010+ only.

Each decade page shows: top-left KDE heatmap, top-right contours with
raw points (blue) and typology points (orange) distinguished,
bottom-left source composition, bottom-right marginals.

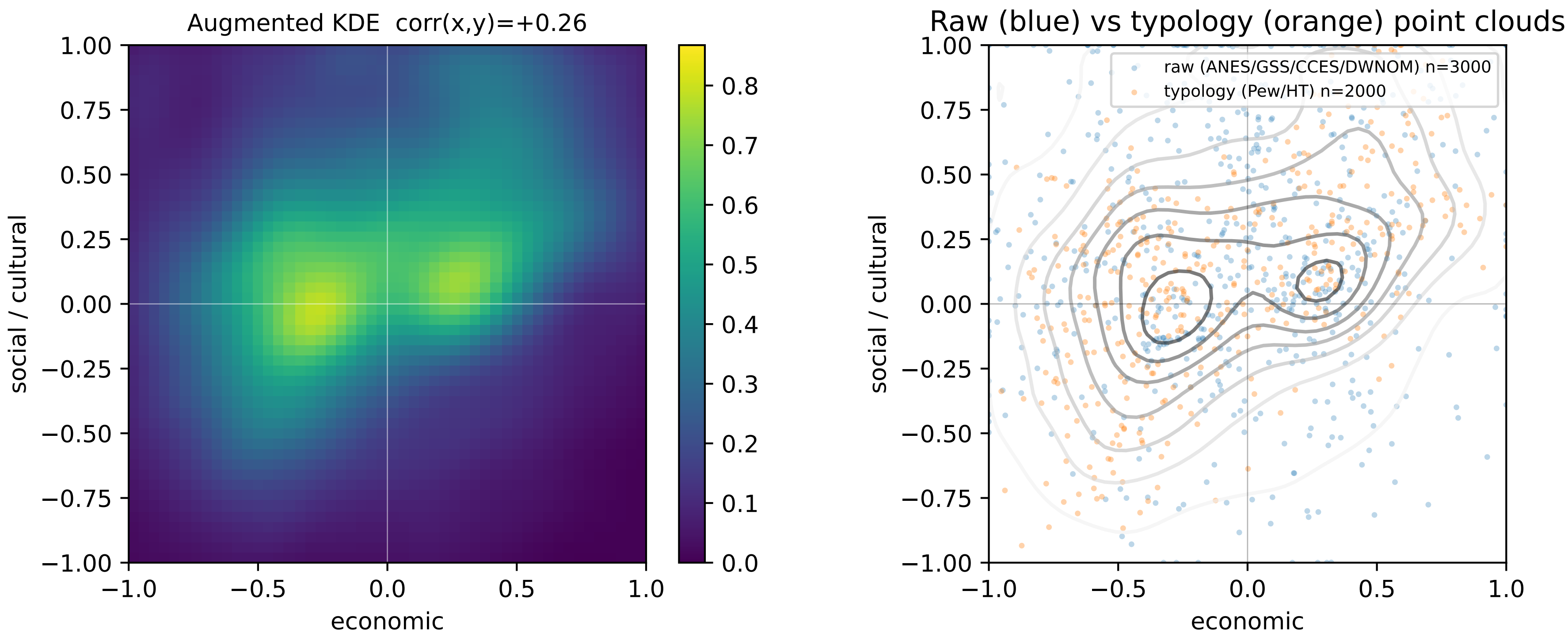
Comparison strip page shows pre-augmentation (Pew-only) vs post-augmentation KDE.
Correlation trajectory plot shows the artifact dilution per decade.

Axes: x = economic (-1 redistributive .. +1 laissez-faire),
y = social/cultural (-1 progressive/secular .. +1 traditional/restrictive).

Methodology: phase9_empirical_targets.md
Augmentation notes: augmentation_notes.md
Raw .npy files: phase9_data/phase9_empirical_kde_<decade>.npy
Pre-augmentation snapshots: phase9_data/phase9_empirical_kde_<decade>_preaug.npy

Generated by phase9_data/render_visualization.py.

Decade 1980 -- 1 typology + 3 raw source(s), 2000 typ pts + 3000 raw pts = 5000 combined (raw weight = 60%)

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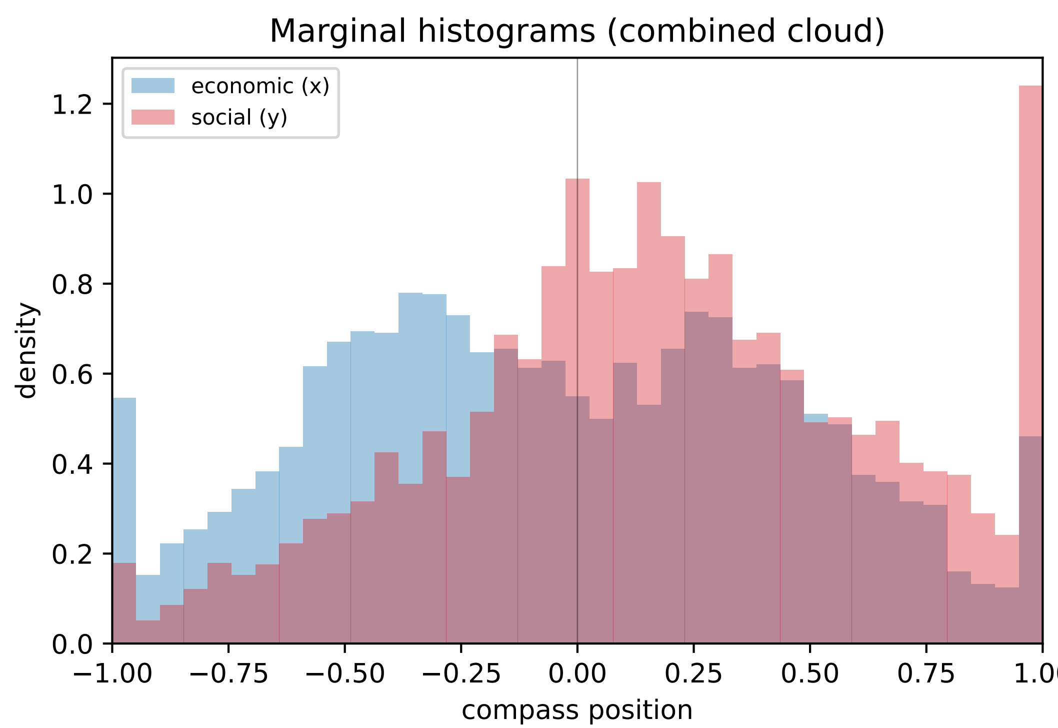
DECADE 1980 composition (raw weight = 60%):

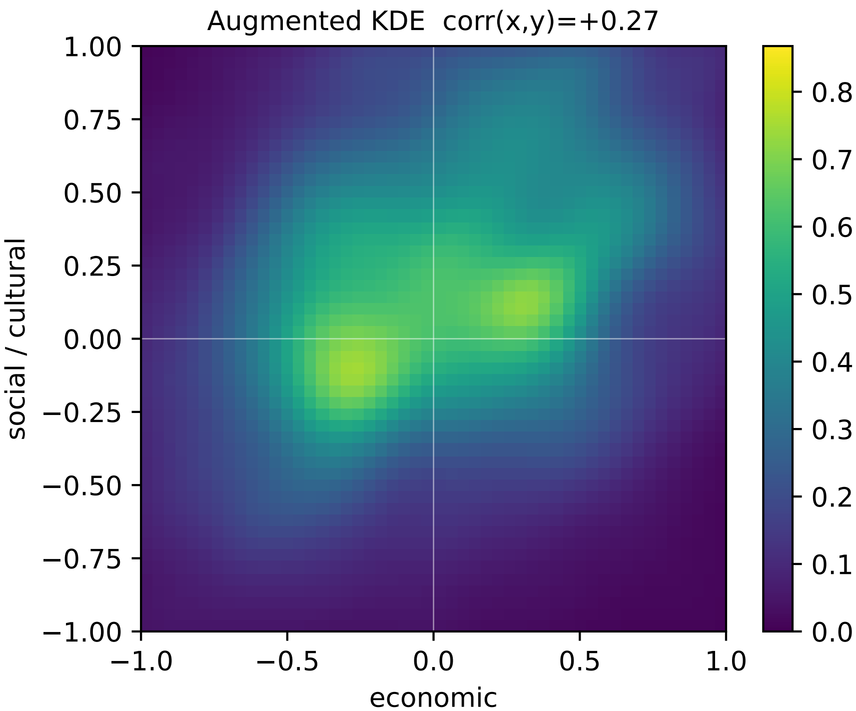
--- Typology sources ---
  Pew_1987  (SD=0.18, downsampled to 2000 total typ pts)

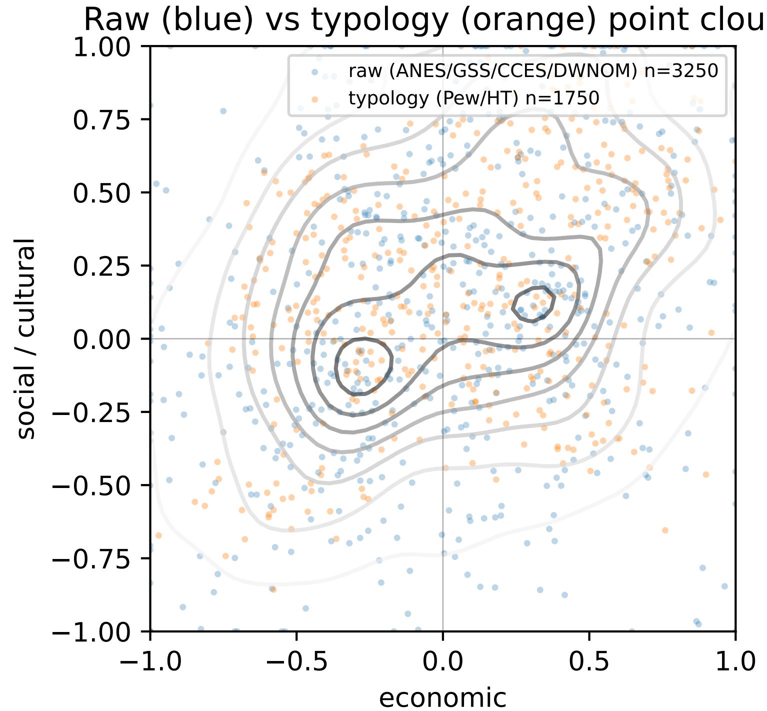
--- Raw-style sources ---
  ANES_1980_synth         within-raw weight=0.55  n=1650
  GSS_1980_synth          within-raw weight=0.35  n=1050
  DWNOM_1980              within-raw weight=0.10  n=300

--- Diagnostics ---
  combined corr(x,y) = +0.2622
  raw-only  corr(x,y) = +0.0988
  typ-only  corr(x,y) = +0.5814
  combined var(x) = 0.244
  combined var(y) = 0.215
  combined mean|x| = 0.419
```



Decade 1990 -- 2 typology + 3 raw source(s), 1750 typ pts + 3250 raw pts = 5000 combined (raw weight = 65%)



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DECADE 1990 composition (raw weight = 65%):

--- Typology sources ---
  Pew_1994  (SD=0.15, downsampled to 1750 total typ pts)
  Pew_1999  (SD=0.15, downsampled to 1750 total typ pts)

--- Raw-style sources ---
  ANES_1990_synth         within-raw weight=0.50  n=1625
  GSS_1990_synth          within-raw weight=0.40  n=1300
  DWNOM_1990              within-raw weight=0.10  n=325

--- Diagnostics ---
  combined corr(x,y) = +0.2734
  raw-only  corr(x,y) = +0.2234
  typ-only  corr(x,y) = +0.4003
  combined var(x) = 0.234
  combined var(y) = 0.218
  combined mean|x| = 0.405
```
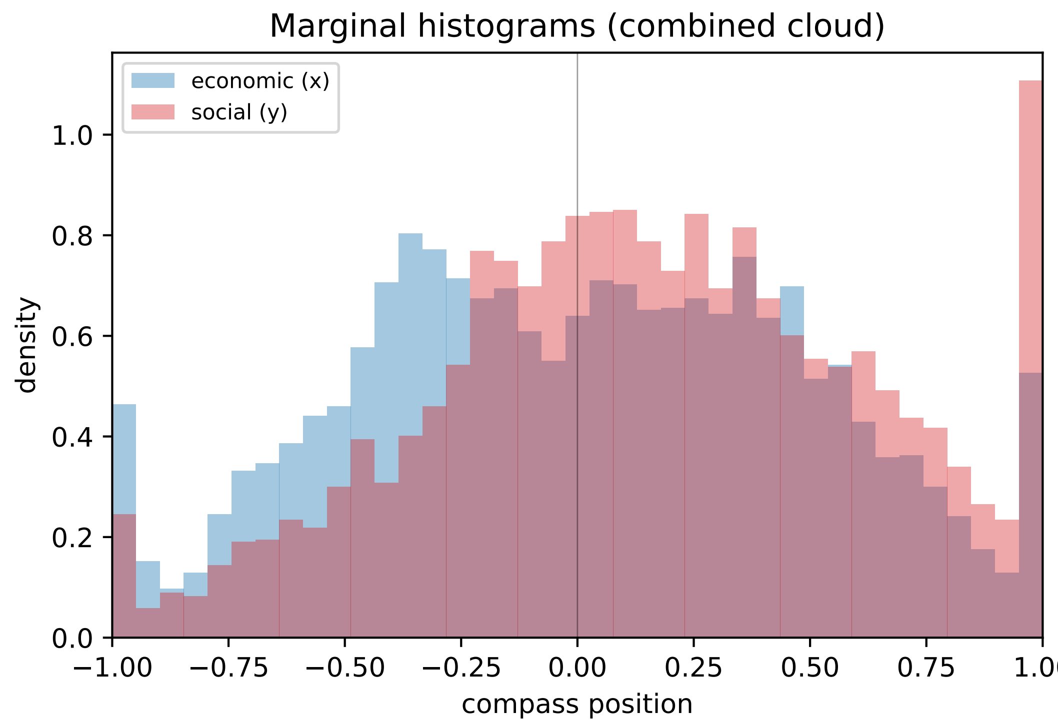
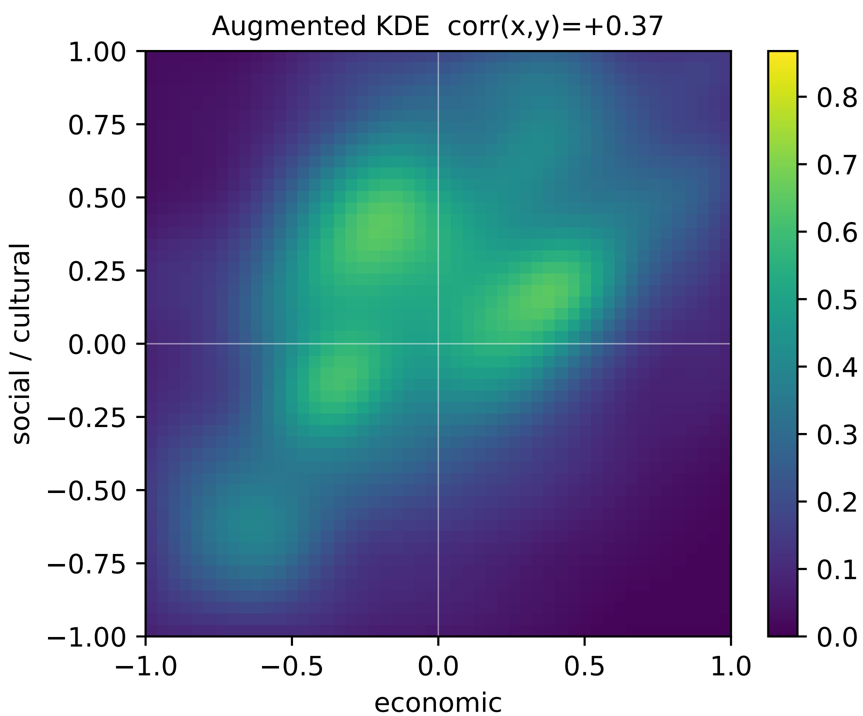
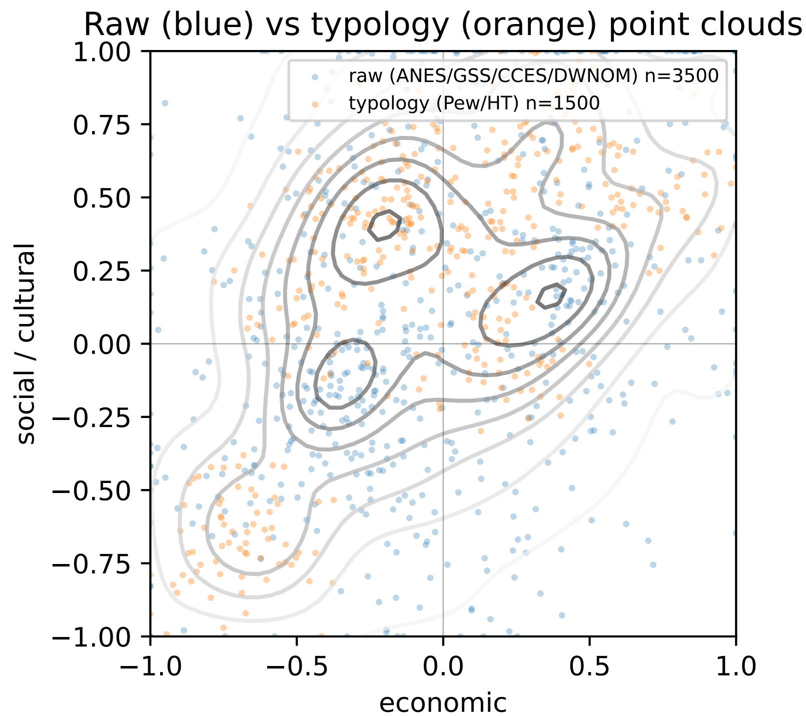



Decade 2000 -- 1 typology + 3 raw source(s), 1500 typ pts + 3500 raw pts = 5000 combined (raw weight = 70%)



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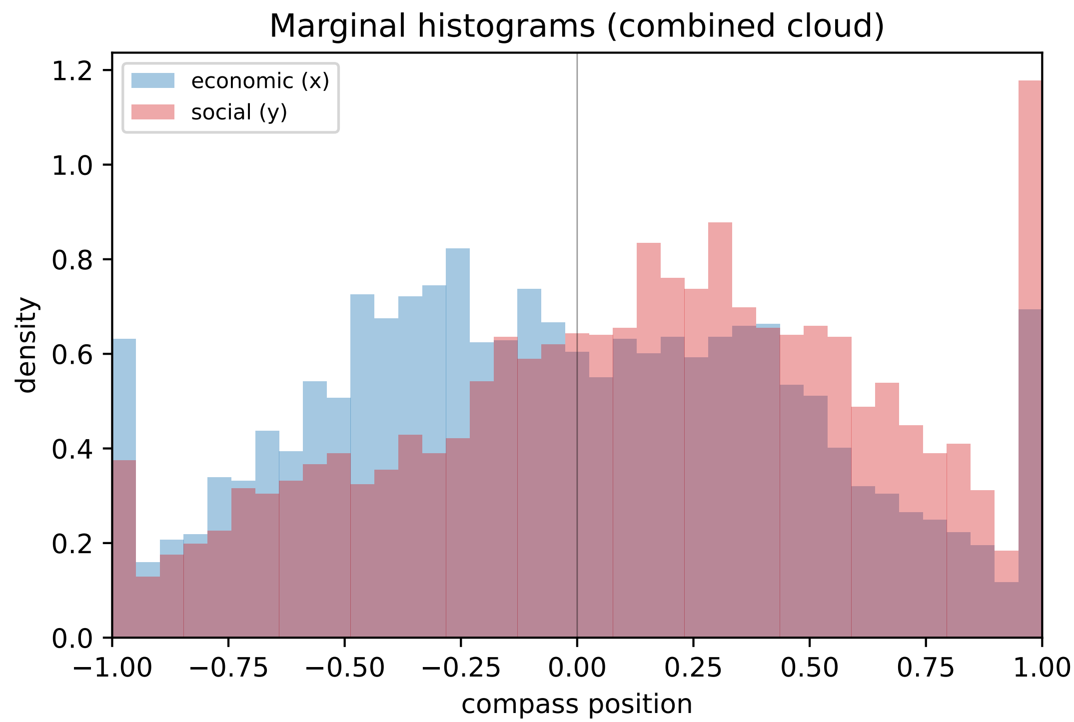
DECADE 2000 composition (raw weight = 70%):

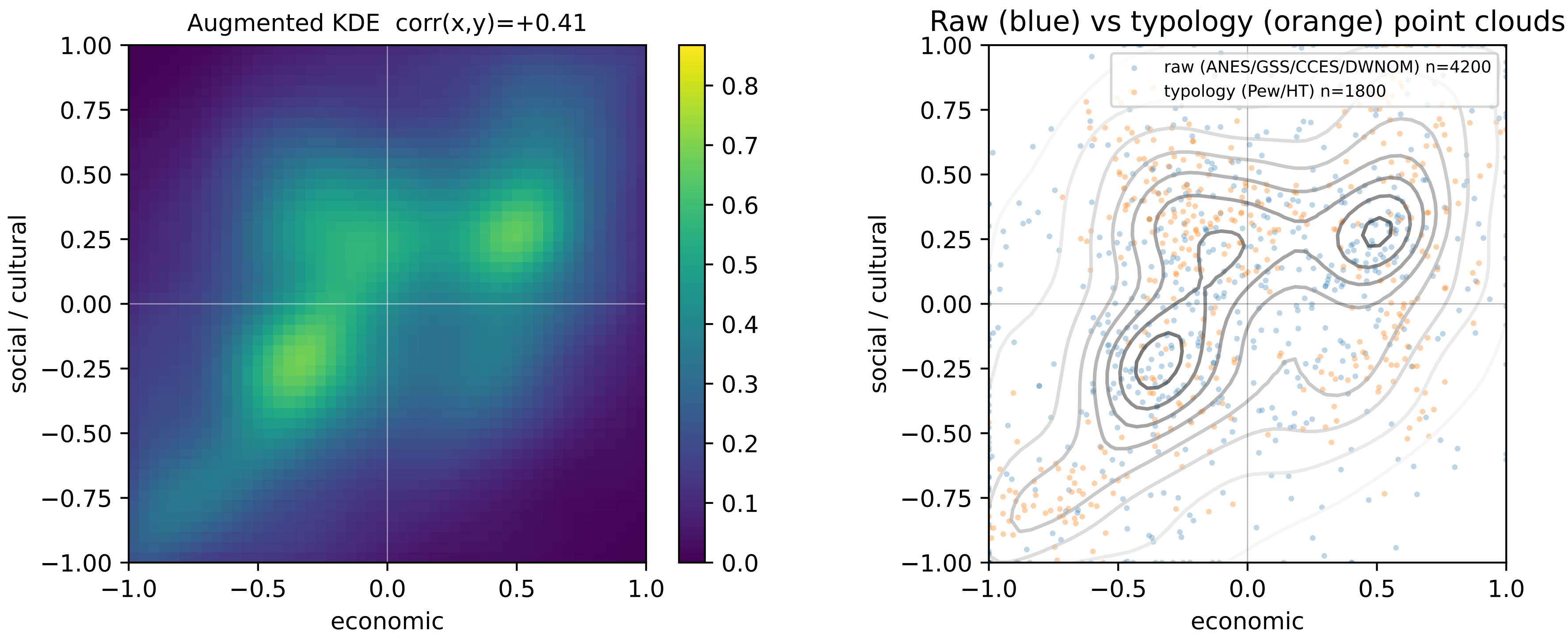
--- Typology sources ---
  Pew_2005  (SD=0.15, downsampled to 1500 total typ pts)

--- Raw-style sources ---
  ANES_2000_synth         within-raw weight=0.50  n=1750
  GSS_2000_synth          within-raw weight=0.40  n=1400
  DWNOM_2000              within-raw weight=0.10  n=350

--- Diagnostics ---
  combined corr(x,y) = +0.3722
  raw-only  corr(x,y) = +0.2894
  typ-only  corr(x,y) = +0.6191
  combined var(x) = 0.254
  combined var(y) = 0.261
  combined mean|x| = 0.421
```



Decade 2010 -- 2 typology + 4 raw source(s), 1800 typ pts + 4200 raw pts = 6000 combined (raw weight = 70%)

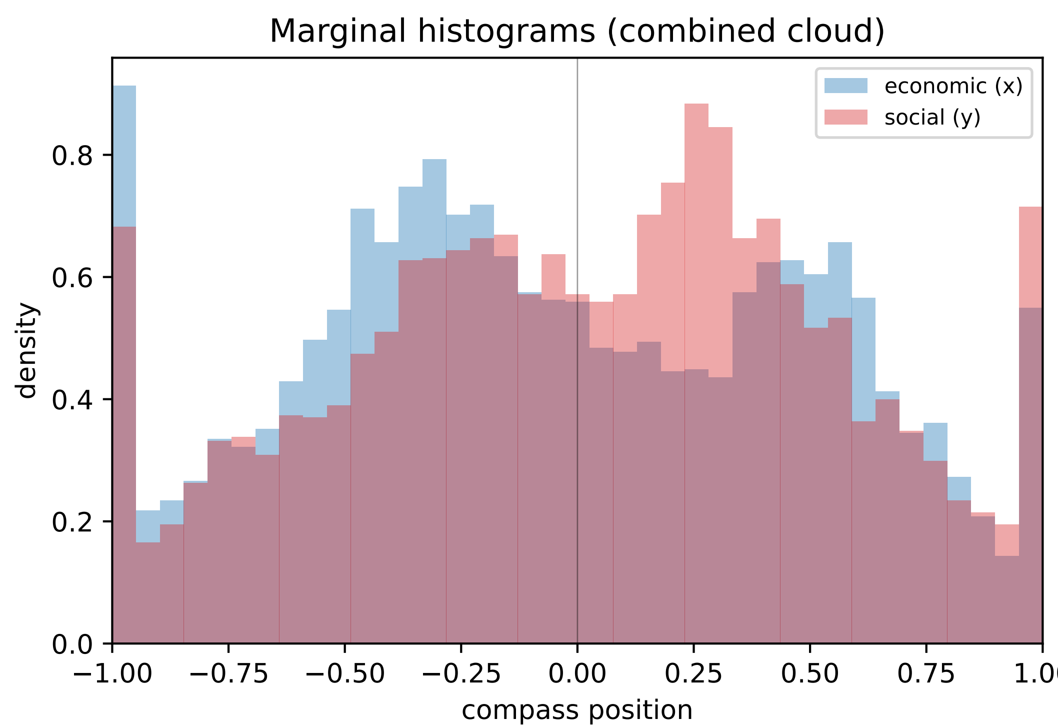



```
DECADE 2010 composition (raw weight = 70%):

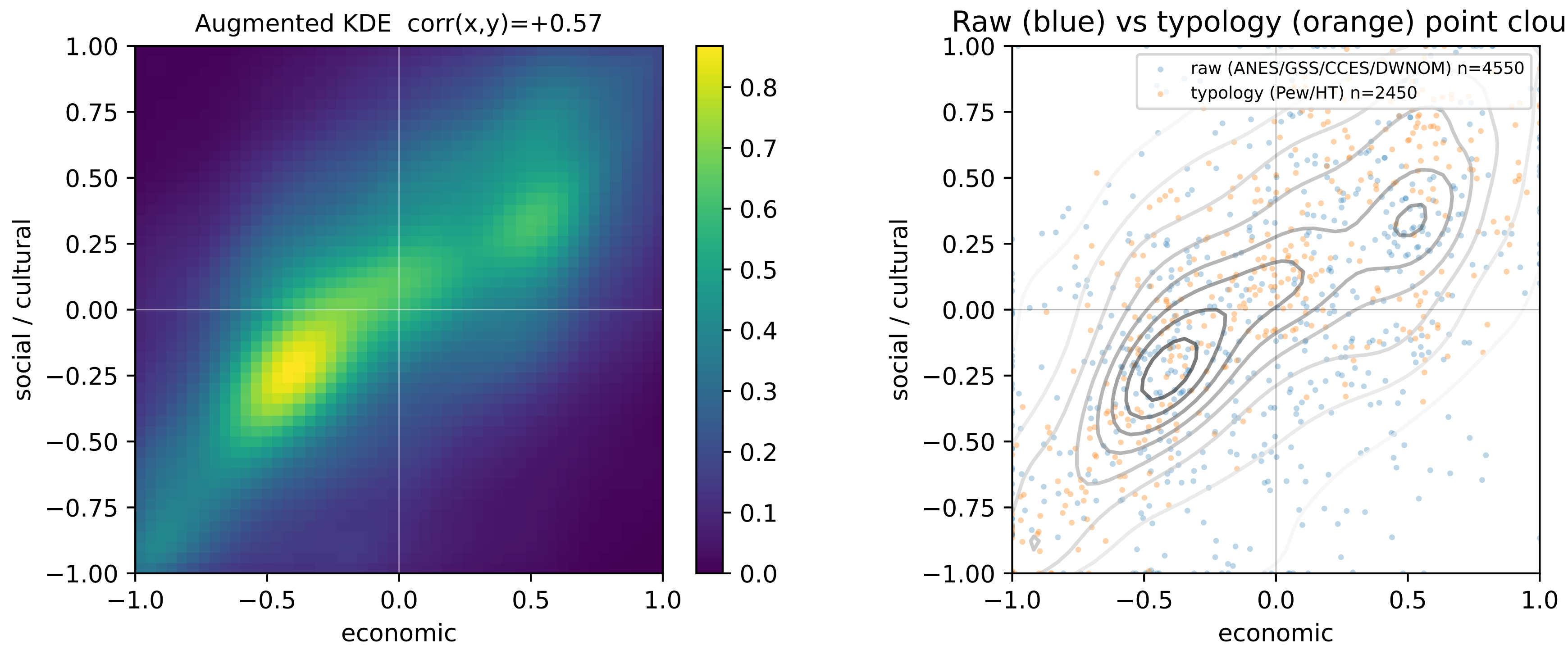
--- Typology sources ---
  Pew_2011  (SD=0.15, downsampled to 1800 total typ pts)
  Pew_2014  (SD=0.15, downsampled to 1800 total typ pts)

--- Raw-style sources ---
  ANES_2010_synth        within-raw weight=0.35  n=1470
  GSS_2010_synth         within-raw weight=0.25  n=1050
  CCES_2010_synth        within-raw weight=0.30  n=1260
  DWNOM_2010             within-raw weight=0.10  n=420

--- Diagnostics ---
  combined corr(x,y) = +0.4147
  raw-only  corr(x,y) = +0.3892
  typ-only  corr(x,y) = +0.4749
  combined var(x) = 0.284
  combined var(y) = 0.262
  combined mean|x| = 0.454
```



Decade 2020 -- 3 typology + 4 raw source(s), 2450 typ pts + 4550 raw pts = 7000 combined (raw weight = 65%)

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DECADE 2020 composition (raw weight = 65%):

--- Typology sources ---
  Pew_2017  (SD=0.15, downsampled to 2450 total typ pts)
  Pew_2021  (SD=0.15, downsampled to 2450 total typ pts)
  HiddenTribes_2018  (SD=0.15, downsampled to 2450 total typ pts)

--- Raw-style sources ---
  ANES_2020_synth         within-raw weight=0.30  n=1365
  GSS_2020_synth          within-raw weight=0.25  n=1137
  CCES_2020_synth         within-raw weight=0.35  n=1592
  DWNOM_2020              within-raw weight=0.10  n=456

--- Diagnostics ---
  combined corr(x,y) = +0.5678
  raw-only  corr(x,y) = +0.4746
  typ-only  corr(x,y) = +0.7662
  combined var(x) = 0.293
  combined var(y) = 0.273
  combined mean|x| = 0.462
```
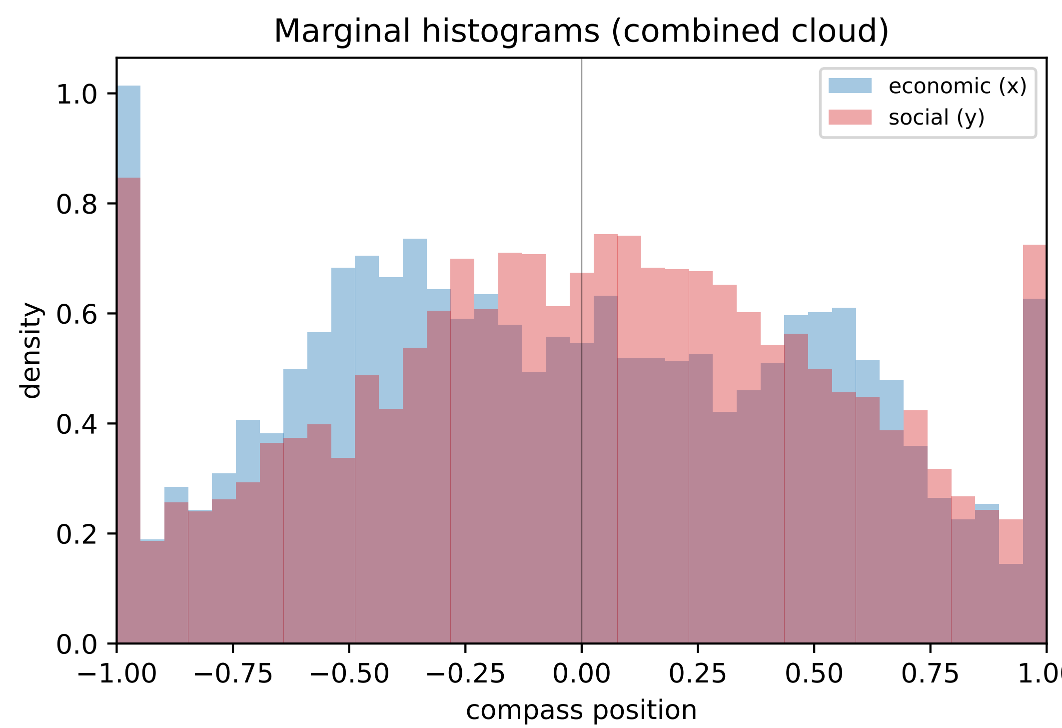
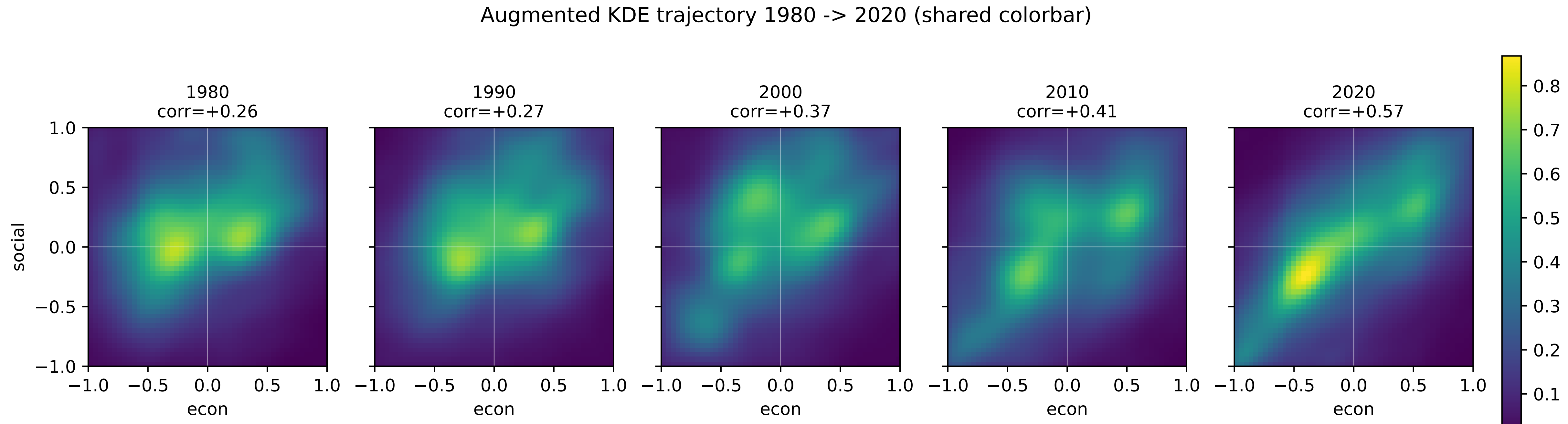
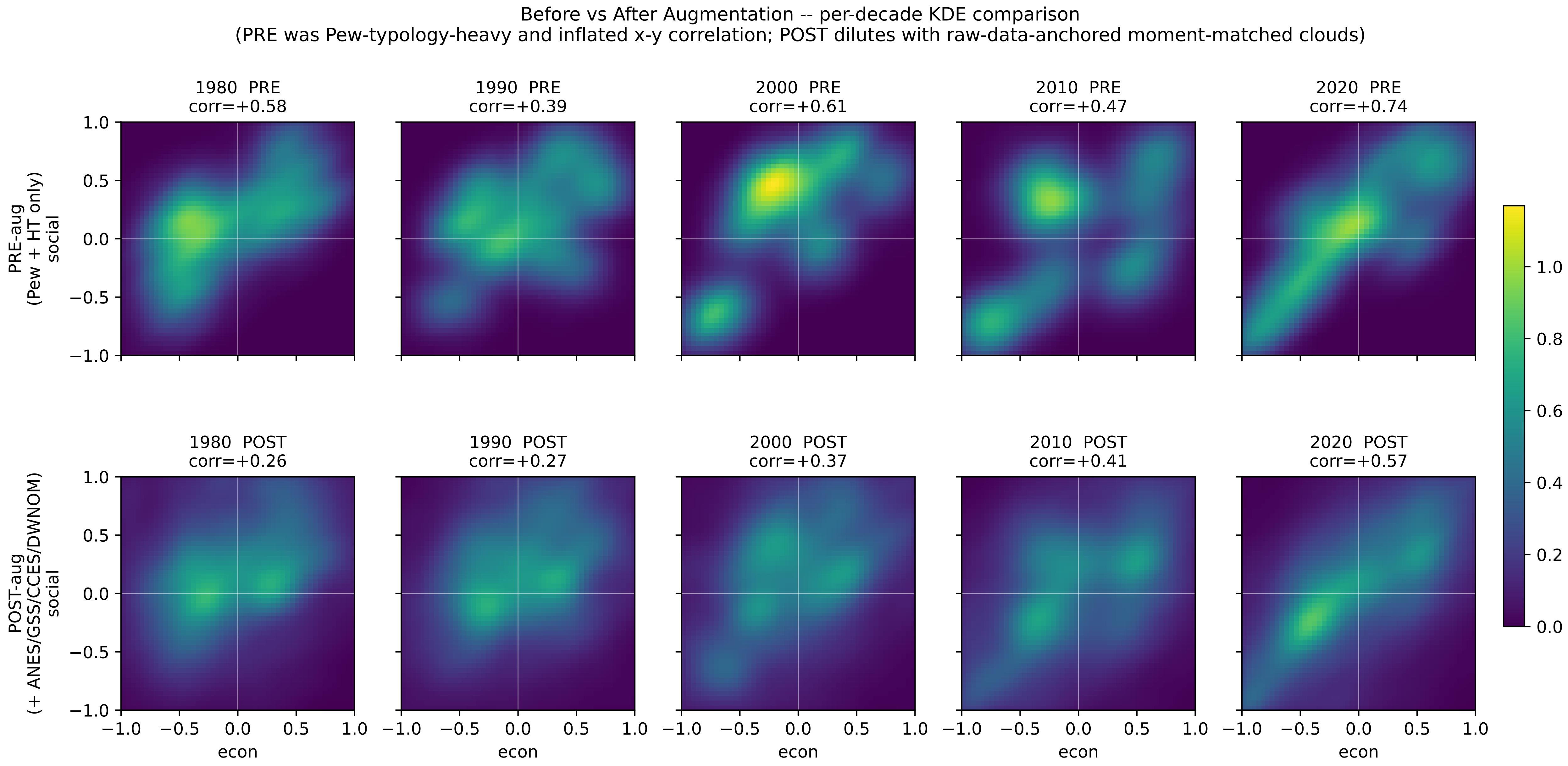
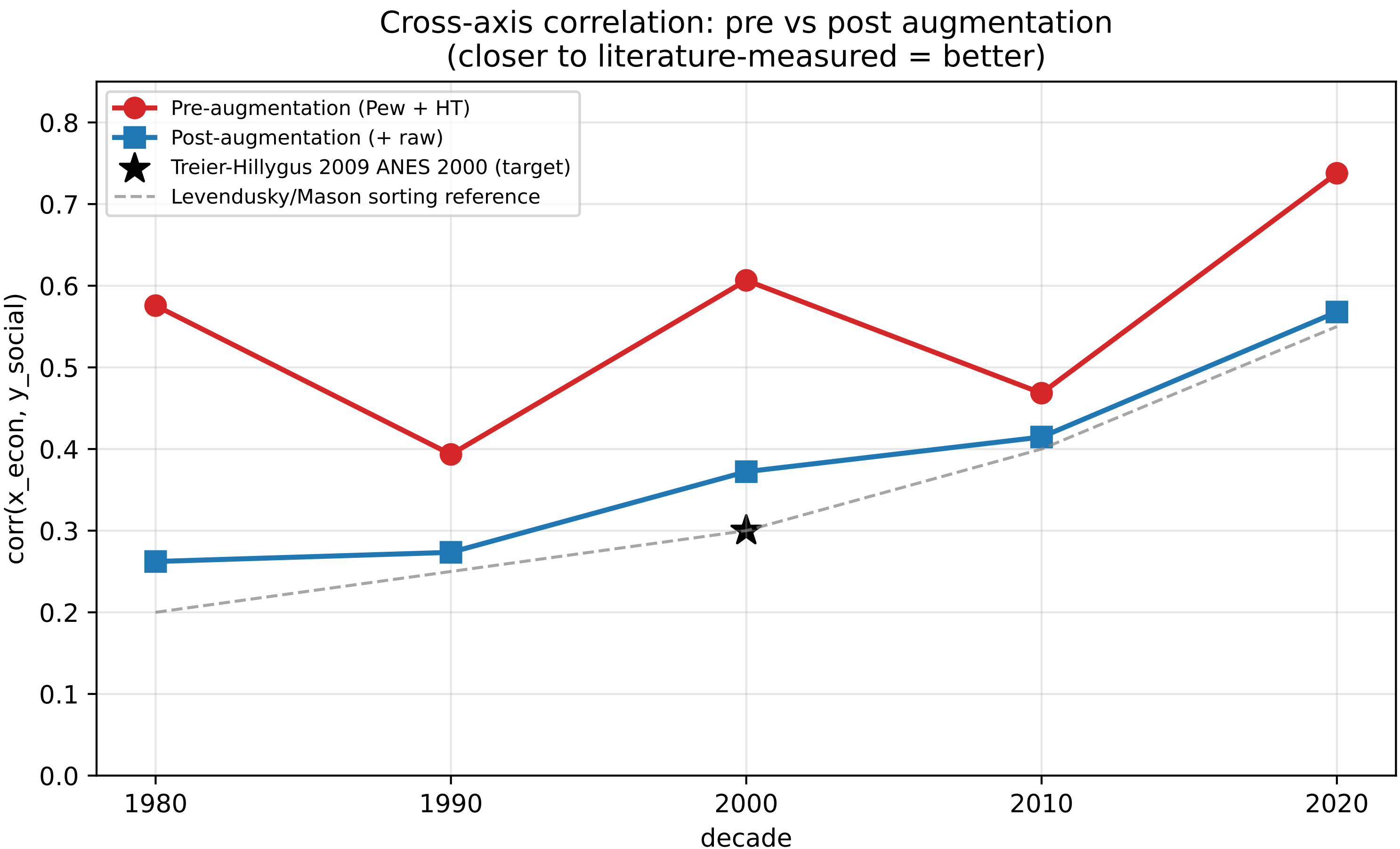




Augmented KDE trajectory 1980 -> 2020 (shared colorbar)
1980
corr=+0.26
1990
corr=+0.27
2000
corr=+0.37
2010
corr=+0.41
2020
corr=+0.57
social
econ
1.0
0.5
0.0
−0.5
−1.0
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0


Before vs After Augmentation -- per-decade KDE comparison
(PRE was Pew-typology-heavy and inflated x-y correlation; POST dilutes with raw-data-anchored moment-matched clouds)
1980 PRE
corr=+0.58
1990 PRE
corr=+0.39
2000 PRE
corr=+0.61
2010 PRE
corr=+0.47
2020 PRE
corr=+0.74
1980 POST
corr=+0.26
1990 POST
corr=+0.27
2000 POST
corr=+0.37
2010 POST
corr=+0.41
2020 POST
corr=+0.57
PRE-aug
(Pew + HT only)
social
POST-aug
(+ ANES/GSS/CCES/DWNOM)
social
econ
−1.0
−0.5
0.0
0.5
1.0
0.0
0.2
0.4
0.6
0.8
1.0



Cross-axis correlation: pre vs post augmentation
(closer to literature-measured = better)
Pre-augmentation (Pew + HT)
Post-augmentation (+ raw)
Treier-Hillygus 2009 ANES 2000 (target)
Levendusky/Mason sorting reference
corr(x_econ, y_social)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
1980
1990
2000
2010
2020
decade